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Remagnetization in southeast China and the collision and suturing of the Huanan and Yangtze Blocks

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ABSTRACT

Nearly all paleomagnetic studies in south China have encountered problems with remagnetization. The region underwent widespread tectonic deformation during the Mesozoic and the warm, humid climate causes rapid weathering of surficial sediments. Paleomagnetic results from three Carboniferous formations from three provinces in southeast China reveal two possible overprinting ages: one is probably a post-folding Cretaceous remagnetization and the other appears to be a Jurassic overprint which may be associated with the collision and suturing of the Yangtze and the Huanan or North China Blocks. The Jurassic overprint is found in sediments with both magnetite and hematite as primary magnetic remanence carriers. The remagnetization pre-dates deformation in at least one of the formations, further constraining the overprint to the Jurassic or late Triassic.

1. Introduction

Recent tectonic studies in south China have shown that the South China Block may be divided into two smaller blocks—the Yangtze Block and the Huanan Block (Fig. 1) [1,2,3]. The Yangtze Block, which is composed of the Sichuan foreland basin and the Yangtze deformed belt, comprises a single tectonic plate. The deformed belt is made up of detached and thrust sedimentary cover with folding as young as early Cretaceous.

The Huanan Block (formerly the South China Fold Belt) is a strongly deformed tectonic block lying to the southeast of the Yangtze Block. Large-scale folds and thrusts are common throughout the block, as are granitic intrusions. Deformation of sediments as young as early Cretaceous is seen in some areas and these deformed sediments are overlain by Cretaceous red beds.

At the contact between the Yangtze and Huanan Blocks, a collision melange has been discovered, which contains metamorphosed flysch of the Banxi Group as well as ophiolites [2]. These rocks were thought to be Precambrian basement [4,5]. Hsü et al. [2,6] presented evidence to support the postulate that at least some of what is called the Banxi Group belongs to a collision melange created during the initial collision and suturing of the two blocks in the late Triassic–Jurassic.

Paleomagnetic studies have been conducted in an attempt to unravel the tectonic history of south China, but nearly all have concentrated on the Yangtze Block [e.g. 7–11]. Few data are available from the Huanan Block. Recently, however, new data have indicated that the postulate of the Hsü group is supported by paleomagnetic results [12,13]. In nearly all of these studies problems with Mesozoic and recent remagnetizations have been encountered.

Paleomagnetic studies of three Carboniferous formations from south China were undertaken as part of a tectonic study of the south China region. The samples were taken from outcrops exposed in Zhejiang, Fujian and Hunan provinces on the Huanan Block in order to test the postulate of Hsü et al. [2] that the Huanan and Yangtze

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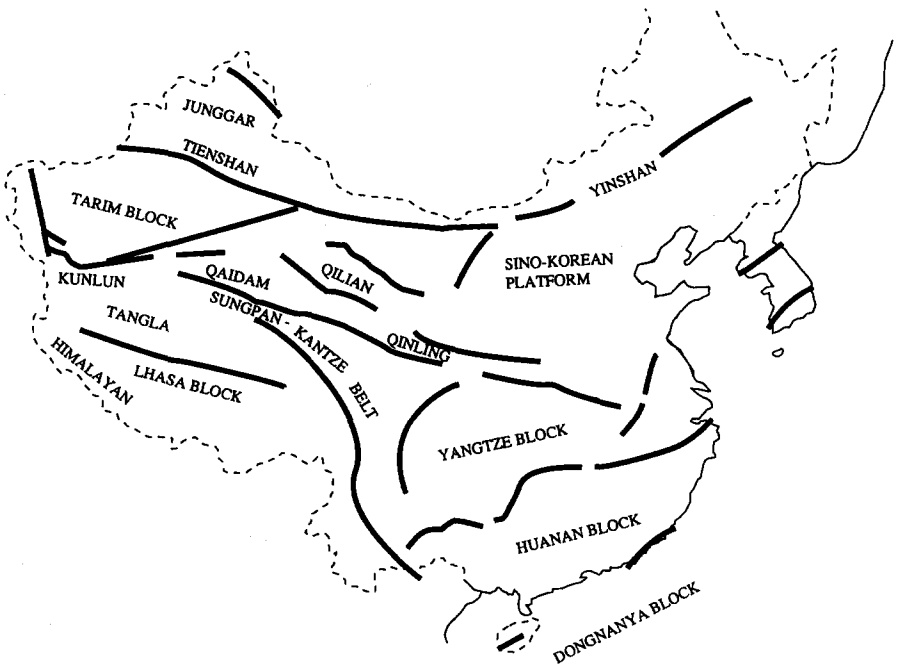


Fig. 1. Tectonic plates of China [after 3]. The Sino-Korean Platform is equivalent to the North China Block and the Huanan and Yangtze Blocks together make up the South China Block.

Blocks were separated during the Paleozoic and the collision and suturing of the two blocks took place in the early Mesozoic (Fig. 2).

2. Sampling sites

The Lower Carboniferous (C_1) Gelaohe Formation was sampled at two sites from two different quarries near Shaoyang City and the village of Dutouqiao (27.2°N, 111.5°E) in Hunan Province. This formation is a dark, argillaceous limestone with shale interbeds bearing brachiopods, rugose corals and conodonts, and it may actually span from the Upper Devonian into the Carboniferous [14]. The first site had a bedding attitude of 008°/33.5° (azimuth/dip), while the second site had a bedding attitude of 327°/63.5°. Both locations are being actively quarried and both sites appeared fresh.

The Upper Carboniferous (C_3) Huanglong Formation was sampled at five sites outside Hangzhou near the village of Zhuji (29.7°N, 120.2°E) in Zhejiang Province, and is correlated with the Maping Stage [14]. This is a fusulinid limestone which also contains coral–brachiopod faunas. The bedding attitude was the same for all five sites (070°/13°). These outcrops, which are along road cuttings, appeared fresh but were fractured, and it was difficult to ascertain the bedding in some places because of thick vegetation.

Four sites from the Chuanshan Formation (C_3) were sampled near the village of Shangcao (25.8°N, 117.3°E) in Fujian Province. The Chuanshan also correlates with the Maping Stage (uppermost Carboniferous), and consists mainly of light coloured, fusulinid-bearing limestones similar to the Huanglong Formation. These outcrops, which again are along road cuttings, appeared fresh and the bedding attitudes for the four sites were similar (av. 114.5°/25°).

3. Results

3.1 The Gelaohe Formation

In nearly all of the Zijderveld diagrams from the Gelaohe Formation, a linear, high-tepmpera-

ture segment decays to the origin between the temperatures of 350°C to 475°C, and multi-vectorial decay of NRM is seen in all samples (Fig. 3a). The low-temperature component seen in the orthogonal vector diagrams between 100°C and 200°C is either a present field overprint or a spurious direction, probably resulting from a VRM acquired during storage in the laboratory before measurement.

The acquisition of IRM at room temperature indicates that low-coercivity minerals are present in these samples with little or no influence from high-coercivity minerals (Fig. 3b). The samples became saturated at fields of about 0.2–0.4 T, which is too high to be the result of the presence of only magnetite. These values are more consistent with pyrrhotite. However, thermal demagnetization of the IRM shows no evidence of

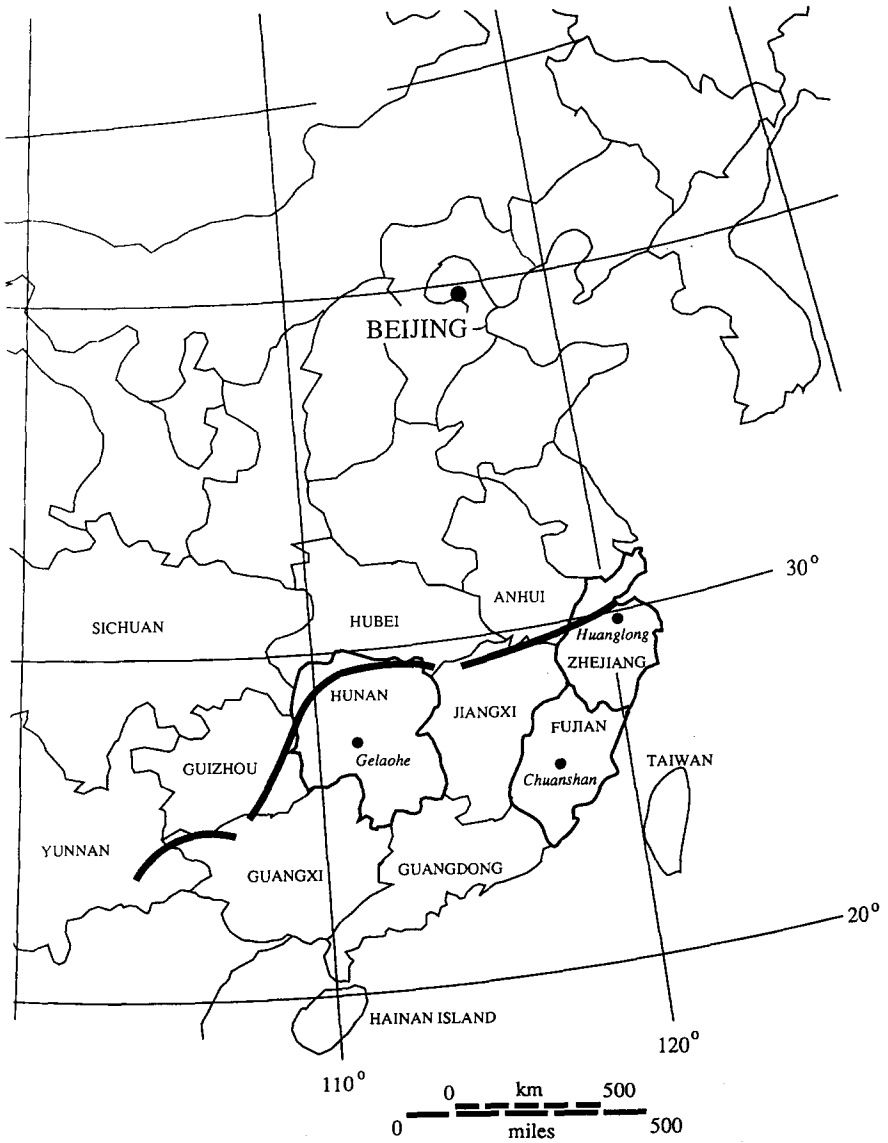


Fig. 2. Map of southeast China showing the sampling sites for the three formations studied. The suture between the Huanan and Yangtze Blocks is represented by the heavy line.

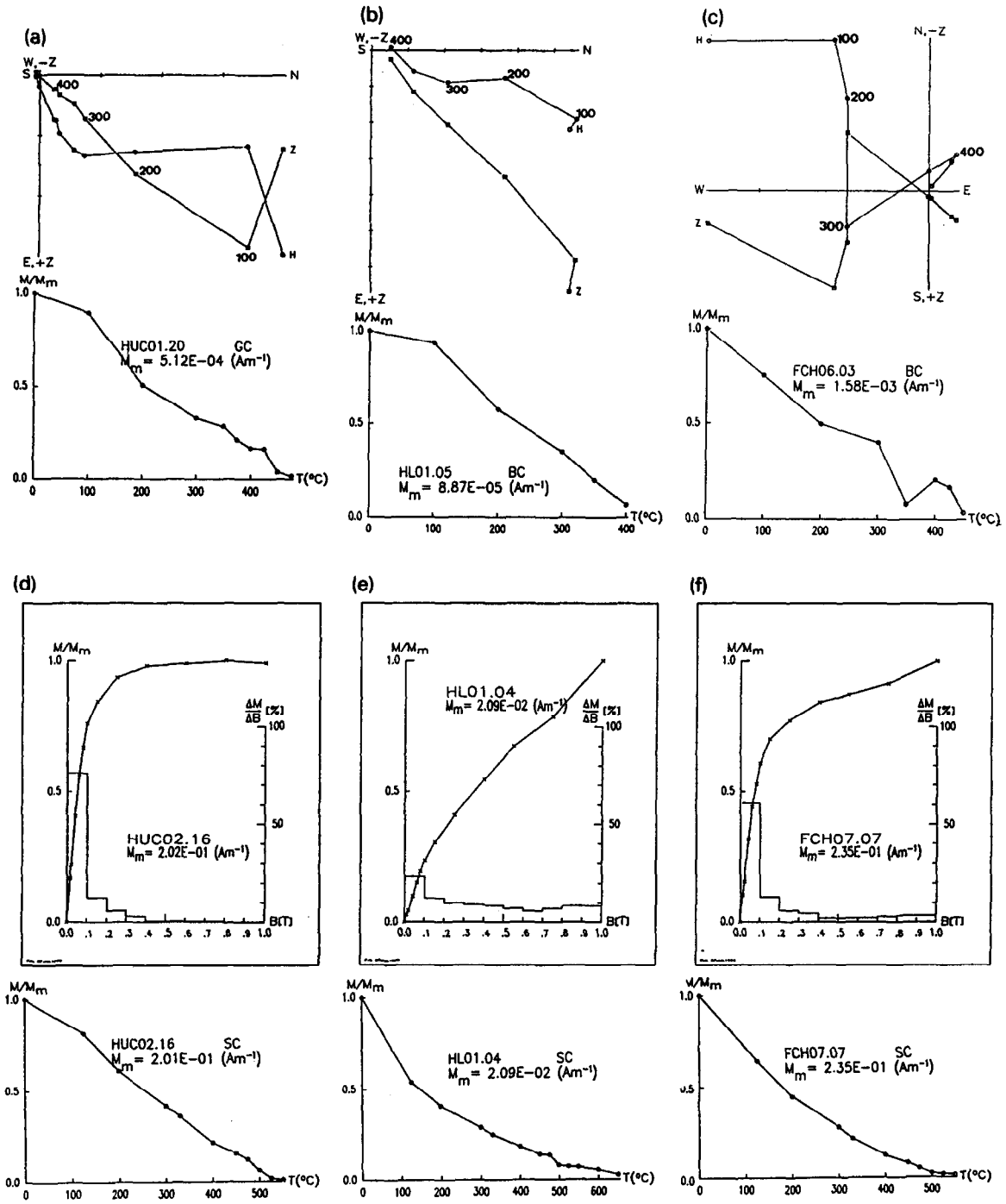


Fig. 3. (a) Thermal demagnetization of a specimen from the Gelaohe Formation. (b) IRM acquisition and thermal demagnetization curves for the Gelaohe Formation. (c) Thermal demagnetization of a specimen from the Huanglong Formation. (d) IRM acquisition and thermal demagnetization curves for the Huanglong Formation. (e) Thermal demagnetization of a specimen from the Chuanshan Formation. (f) IRM acquisition and thermal demagnetization curves for the Chuanshan Formation.

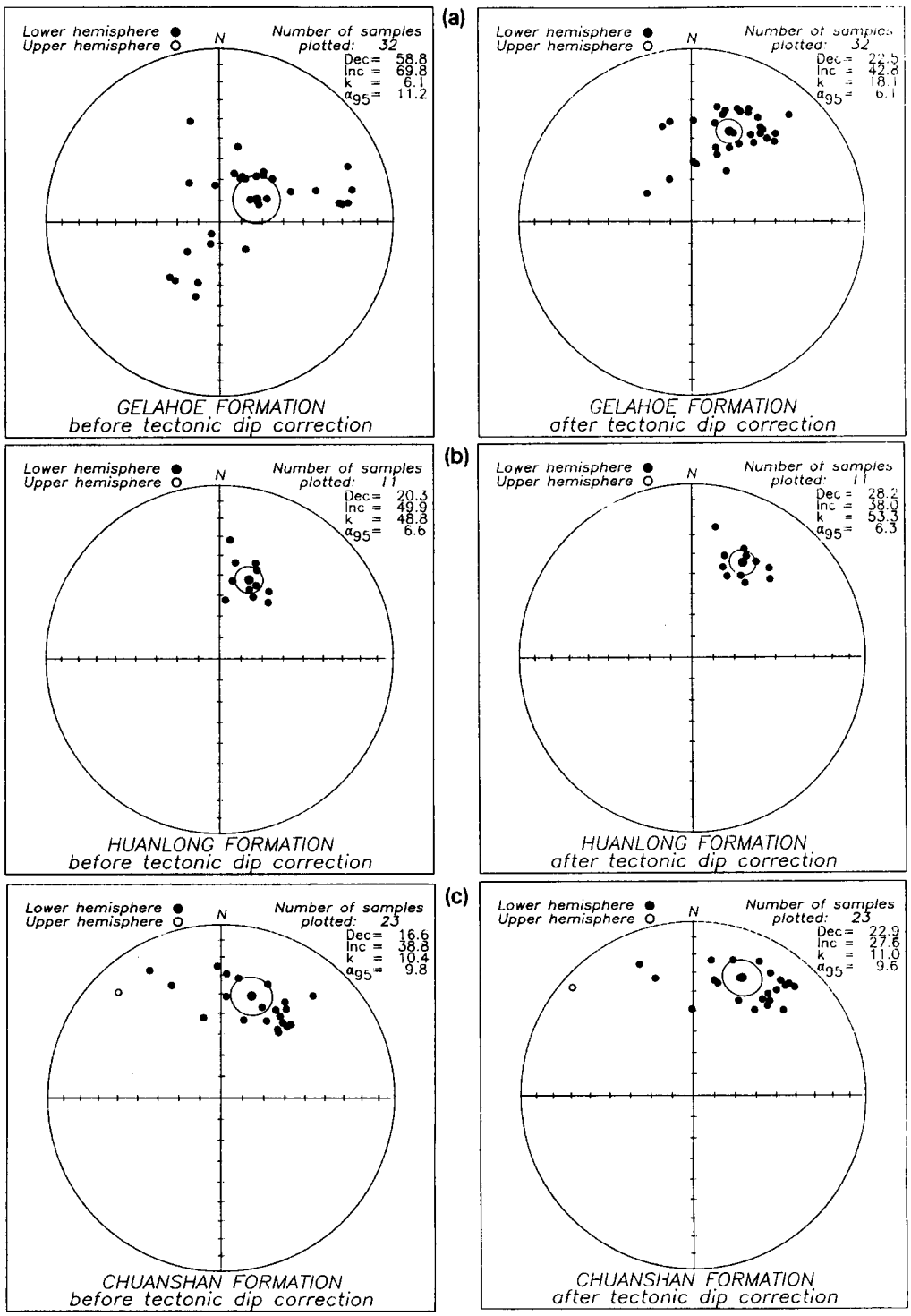


Fig. 4. (a) Equal-area projection of the characteristitc remanence directions for the Gelaohe Formation. (b) Equal-area projection of the characteristic remanence directions for the Huanglong Formation. (c) Equal-area projection of the characteristic remanence directions for the Chuanshan Formation.

pyrrhotite in these samples. This being the case, the high-temperature directions isolated at temperatures above 320°C are presumed to be carried by magnetite.

Linear segments from the orthogonal vector diagrams were analyzed using the least squares methods of Kirschvink [15]. Stable components were isolated for 32 samples from this formation and the three-fold increase in Fisher's precision parameter k [16] after application of a bedding correction indicates that the magnetization was acquired before folding of the beds (Fig. 4a). The *in-situ* sample directions are dispersed and do not preserve a recent overprint.

The mean declination of the sites is 22.5° and the inclination is 42.8° after tectonic correction. This corrected mean direction yields a paleopole which coincides closely with Jurassic paleopoles for the Yangtze Block [17], indicating that these sites from the Huanan Block were probably remagnetized at that time. The Mesozoic part of the Yangtze APWP, however, is not well defined, and the ambiguity of the pole path makes a precise determination of the remagnetization age difficult. The fact that the sites pass an attitude test is consistent with geological evidence that folding took place as late as Cretaceous in many parts of south China. This implies that the secondary overprint was acquired before folding of the beds and the remagnetization age is therefore constrained to older than early Cretaceous (the age of the youngest folds).

3.2 The Huanglong Formation

Twenty-four samples from the Huanglong Formation were thermally demagnetized, and stable, high-temperature directions were resolved in 11 of these 24 samples (Fig. 3c). The remainder exhibited unstable behaviour during demagnetization and no stable high-temperature directions could be isolated.

Only one component (in some cases two) of magnetization is seen in the orthogonal vector plots. The unblocking temperatures for these samples are quite low. Most samples lose their measurable remanence by 400°C. In some cases measureable intensity remains above these temperatures. However, at higher temperatures an increase in intensity makes it impossible to isolate any stable component. This behaviour is reflected in the bulk susceptibility measurements, which show an increase beginning above 300°C. Apparently, new ferromagnetic minerals are forming during the heating process in significant amounts, and acquired a remanence either in the cooling chamber or when they were brought out for measurement. Alternating field demagnetization was ineffective at isolating a stable component of magnetization, even after heating to 150°C, indicating that hematite may be present in the samples, although the blocking temperatures appear quite low.

Isothermal remanent acquisition curves indicate that the remanence is dominated by high-coercivity minerals, with only a small contribution from minerals with low coercivity (Fig. 3d). Saturation is not reached by 1.0 T, and a nearly linear increase in intensity is seen in the curves above 0.25 T. Thermal demagnetization of the IRM shows that measurable intensity persists above 600°C. It is likely that the stable component isolated in these samples is carried by hematite with low blocking temperatures.

The paleopole position derived from the mean *in-situ* direction (Fig. 4b) falls near the Jurassic segment of the APWP for the Yangtze Block. Due to the uncertainty in the Yangtze Block APWP, as previously mentioned, however, this direction could also represent a Cretaceous remagnetization. This will be discussed further in the next section. Upon correction for bedding attitude the pole moves farther off of the APWP, although the direction is still consistent with a Jurassic remagnetization. As no fold or attitude test is available, it cannot be determined whether the Jurassic or early Cretaceous overprint seen in these sediments pre-dates or post-dates folding, although the post-folding pole is in much better agreement with the APWP.

3.3 The Chuanshan Formation

The Zijderveld diagrams for the Chuanshan Formation reveal multi-vectorial demagnetization paths for all samples (Fig. 3e). An initial component is removed by 100°C, and is probably a laboratory VRM as it is not associated with the present field direction. One or two more components are seen in most of the samples and in 21

TABLE 1

Paleomagnetic results from the Gelaohe, Huanglong and Changxing Formations and comparison with the Jurassic and Cretaceous reference poles for the South China Block from [17]

Formation	Age	Pole position bedding corrected			Pole position *in situ*		
		Lat. N	Lon. E	A_{95}	Lat. N	Lon E.	A_{95}
Gelaohe	C_1	69.7	203.1	6.9	39.9	250.1	16.6
Huanglong	C_3	63.3	221.8	6.2	72.4	201.8	7.3
Chuanshan	C_3	65.8	230.3	9.4	74.3	218.2	9.9
Reference poles for the South China Block from [17]							
	J_1	69.4	192.4	7.4			
	J_2	65.0	186.2	12.0			
	J_3	72.3	208.6	13.5			
	K_1	75.1	214.1	6.2			
	K_2	78.9	207.0	9.0			

of the samples a stable, high-temperature direction was resolved between 400°C and 500°C.

For one site, the NRM intensity was extremely weak and by 100°C no longer measurable. Most of the samples from this formation are demagnetized by 450–500°C. In some samples a large contribution from a reversed NRM component was seen between 300°C and 400°C.

A dominance of low-coercivity minerals is seen in the IRM curves. However, the samples are not completely saturated by 1.0 T, indicating at least partial contribution of high-coercivity minerals to the remanence (Fig. 3f). Demagnetization of IRM shows a steady decay to temperatures of about 500–550°C, probably representing magnetite.

The *in-situ* directions of the 23 samples for which high-temperature stable components were resolved exhibit some scatter (Fig. 4c). The mean direction yields a paleopole which coincides closely with the early Cretaceous pole for the Yangtze Block [17]. After correction for bedding attitude the pole moves to a position off the APWP, but the circle of confidence encompasses the late Jurassic pole. As with the Huanglong Formation, it cannot be determined whether this remagnetization pre-dates or post-dates folding, although the post-folding direction is in much better agreement with the APWP. This indicates that these sites may carry a post-folding Cretaceous overprint, which has been shown to be prevalent in many sediments in southeast China [18], or a pre-folding Jurassic remagnetization as seen in the Gelaohe Formation. Again, however, although the results agree well with the Cretaceous paleopole for Yangtze, a precise age of remagnetization cannot be determined due to the ambiguity of the pole path. The results from these formations are summarized in Table 1 and compared to reference directions.

4. Discussion

The Carboniferous Gelaohe and Huanglong Formations that we have studied in the Huanan Alps appear to preserve paleomagnetic directions which correspond to the Jurassic segment of the Yangtze Block APWP [8,17] (Fig. 5a). The overprint in the Gelaohe Formation is recognized after correction for bedding attitude, indicating that the remagnetization was acquired before the folding, which ended in the early Cretaceous. In the Huanglong Formation this overprint is ambiguous and may have been acquired either before or after folding. It is also possible that the Chuanshan Formation was remagnetized in the Jurassic before folding, but the data agree better with a post-folding Cretaceous remagnetization (Fig. 5b).

Due to the paucity of paleomagnetic data available from the Yangtze Block, the positions of statistically distinct Mesozoic poles are difficult to establish. This is seen in the most recent and comprehensive APWP of [17]—against which the data from this study have been compared. Al-

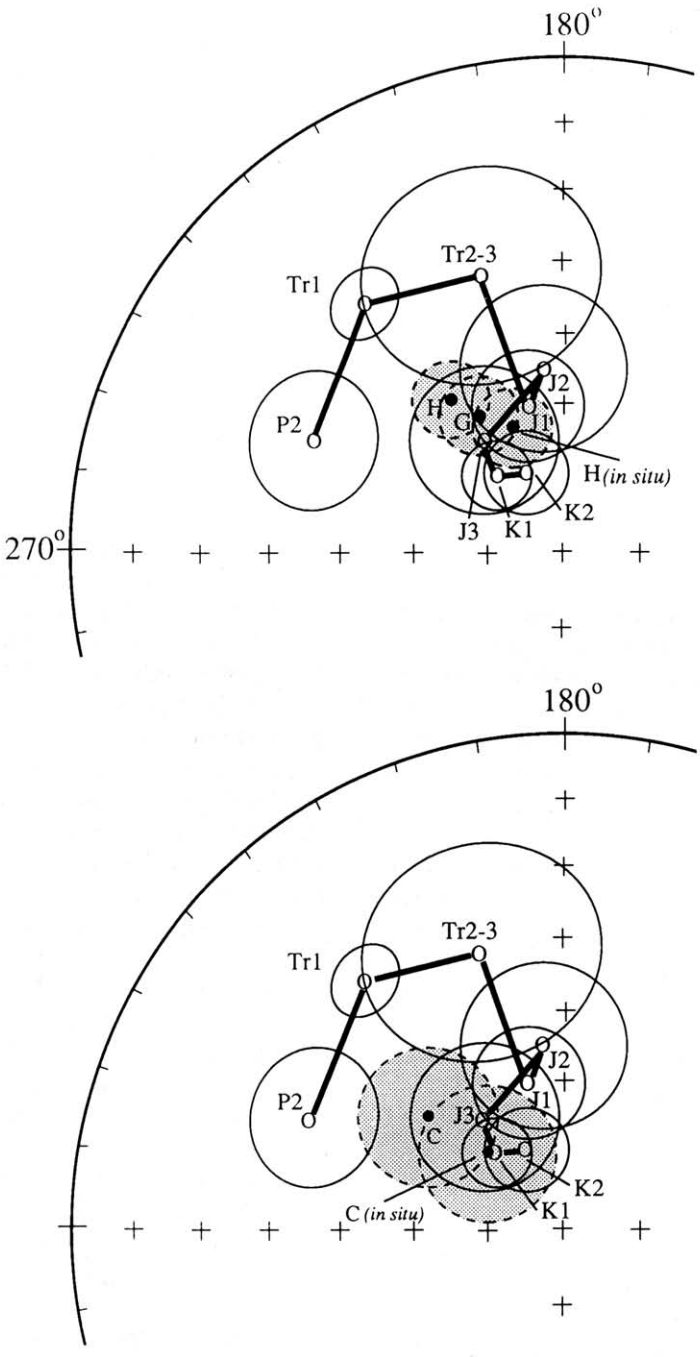


Fig. 5. (a) Apparent polar wander path for South China Block from [17] with A_{95} confidence limits. Dots represent the poles for the Gelaohe (G) and Huanglong (H) Formations after bedding correction. The *in-situ* pole for the Huanglong Formation is also indicated. The A_{95} confidence limits for these formations are shown in grey. (b) Apparent polar wander path for the South China Block from [17] with A_{95} confidence limits. Dots (C) represent the poles for the Chuanshan Formation, with the A_{95} in grey.

though the agreement of the Gelaohe and Huanglong poles with the Jurassic segment of the APWP is very good, the age of remagnetization cannot be well constrained and may be slightly older or younger. The fact that the overprint in the Gelaohe Formation is pre-folding, however, is a strong indication that the remagnetization in that formation can be constrained to the upper Triassic or Jurassic.

It is possible then, that the remagnetization of these Carbonifersous formations from the Huanan Block may have resulted from processes which affected these sediments during the later stages of collision and deformation between the Yangtze and Huanan Blocks. This is supported by geological evidence that collision of the two blocks occurred during the late Triassic but that deformation continued into the early Cretaceous [2]. It should be stressed, however, that this geological evidence is still controversial, and an alternative explanation will be discussed in the next section. The remagnetization of these south China sediments may have an analogy in the Appalachians of the eastern United States.

Widespread remagnetization resulting from the formation of secondary magnetic minerals has been recognized in the eastern U.S. and is thought to be a result of the migration of tectonic fluids during the Alleghanian orogeny [19,20,21]. It has also been shown that as two continental blocks collide, hydrodynamic circulation of groundwater is initiated, which may result in the oxidation of pyrite to magnetite and may also account for the formation of secondary hematite [21–24]. Heat from the warm, saline groundwater is now thought to to be one of the major factors facilitating conversion to ferromagnetic minerals [25,26,27].

In the case of the Appalachians, the acquisition of a thermoviscous remanence (TVRM) has also been considered for remagnetized magnetite-bearing rocks. For comparison, calculations of the stability of a TVRM acquired by magnetite in the middle Jurassic using the thermal activation nomogram of Pullaiah et al. [28], shows that for a laboratory unblocking temperature of 450°C the Gelaohe sediments would have to have been heated to temperatures between 350°C and 400°C during burial to preserve a TVRM with the observed blocking temperatures, assuming that the TVRM was acquired over one

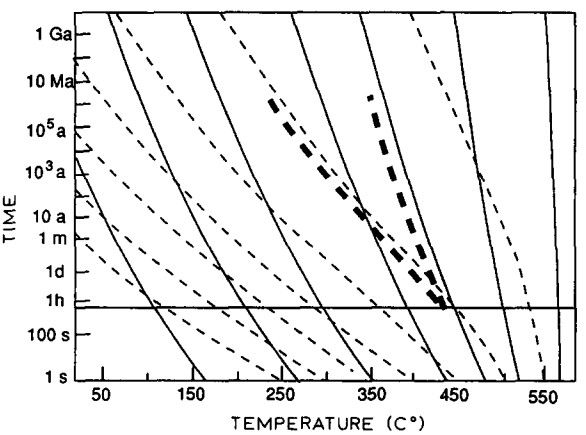


Fig. 6. Thermal activation nomograms for single-domain magnetite. Solid curves are from [28] and dashed curves are from [29]. Heavy dashed curves represent a TVRM component which is demagnetized in the laboratory at 450°C and which was acquired over a duration of one to ten million years.

to ten million years (Fig. 6). These temperatures correspond to a greenschist degree of metamorphism, which is not seen in the sediments.

Recent debate has arisen, however, concerning the ability of the Pullaiah method to accurately predict the behaviour of multi-domain magnetite grains during thermal activation, as their nomogram is constructed based on calculations using Neel's single-domain theory [29,30,31]. Kent [31] demonstrated that the Middleton and Schmidt curves—which were calculated based on Walton's [32] thermal activation theory which assumes a logarithmic grain-size distribution—were more accurate at predicting the burial temperature for the rocks he studied. Using the Middleton and Schmidt thermal activation nomogram, a burial temperature of less than 250°C could result in a TVRM in these sediments. Temperatures of this order are more plausible; however, the thermal maturity of these sediments is not known. The TVRM question, however, remains unresolved, and Jackson [33] has recently shown that Kent's analysis may not sufficiently explain the source of the late Paleozoic overprint in the latter's samples.

A TVRM resulting from the collision and suturing of the two blocks requires that much of the sediment be buried during collision, enough to elevate the temperature of the rocks by approximately 250°C. This would correspond to a burial

depth of over 8 km. The Huanan Block, however —from which the Gelaohe and Huanglong sediments come—was thrust over the Yangtze Block during the initial stages of collision and prolonged deep burial of these sediments is not likely.

5. Conclusions

The paleomagnetic results for the Gelaohe and Huanglong Formations of the Huanan Block indicate that the sediments may have been remagnetized in the Jurassic during suturing of the Huanan and Yangtze Blocks. The pole from the Chuanshan Formation, however, appears to preserve a later remagnetization, although the possibility of a Jurassic remagnetization of this formation cannot be ruled out.

A Jurassic overprint has been recognized in the Emeishan Basalts of Guizhou on the Yangtze Block, although whether the overprint is recognized before or after bedding correction was not reported [8]. These authors [Lin et al., 8] also reported Mesozoic remagnetization of Devonian, Silurian and Ordovician sediments, although a more precisely constrained age of remagnetization could not be given. Permian sediments from the Changxing Permian–Triassic type section studied by Li [34,35] may also preserve a Jurassic remagnetization. Data from these studies indicate that the Jurassic overprint may be widespread in southeast China.

Kent et al. [18] recognized a late Mesozoic to Recent overprint in a large number of their samples from the Yangtze Block. They attributed this remagnetization to Yanshanian orogenic activity. The remagnetization that they report is late Jurassic and younger in age, and was acquired after deformation took place, as is evident from four negative fold tests. This may be the same post-folding overprint that is seen in the Chuanshan Formation on the Huanan Block. It may also be the same overprint as that seen in the Huanglong Formation.

The remagnetized samples from the Gelaohe and Huanglong Formations, although they are taken from two different provinces, are all of the same polarity. This may be an indication that the remagnetization of all of these sediments took place within a relatively short time—being constrained to one polarity interval. It must also be considered that this phenomenon may be evidence of an unusually long-duration normal polarity interval during the Jurassic. This has been postulated before for the middle Jurassic based on marine magnetic anomalies and on studies of land sections [36,37]. If this case is assumed, migration of the tectonically driven fluids over large distances in a short period of time is not required.

The Huanglong and Gelaohe sediments come from near the Huanan–Yangtze suture zone and the remagnetization appears to have occurred during deformation along that zone. This indicates that these sediments may have been remagnetized near the conclusion of, and that the remagnetization was a result of, the collision and suturing of the two blocks during the Jurassic.

For the Jurassic, however, paleomagnetic evidence indicates that the Yangtze Block was also being sutured to the North China Block [8,10,17]. The previously discussed overprint seen in the eastern U.S. can be traced all the way into the mid-continent regions, indicating that sediments far from the suture zone may also be affected by remagnetization processes [25]. The sediments of this study are certainly near enough to the Yangtze–North China Block suture zone to have been remagnetized during the collision if the eastern U.S. situation may be considered an accurate analogy. The possibility that the collision caused the remagnetization seen in this study then, must also be considered, and the question of which collision was responsible for the remagnetization remains open.

Whether the mechanism of remagnetization was only one or a combination of the processes discussed in the last section, the evidence for tectonic activity and block collision in the southeast China region during the Mesozoic is compelling. The identification of the Mesozoic overprint further supports the geological and paleomagnetic evidence that accretion of some of the southeastern blocks of China was occurring during this time.

Acknowledgements

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